

Economic Incentives and Religious Group Membership in Africa

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Why Join Religious Groups?

Benefits of joining religious groups...

- Psychological
 - From religious beliefs (e.g., the afterlife, moral guidance)
 - From social connections (emotional, social support)
- Economic
 - From social networks (information flows, informal insurance)

Costs of joining religious groups...

- Psychological
 - Social pressure for norm adherence
- Economic
 - Financial contributions, tithing
 - Distance (travel cost)
 - Adherence to religious taboos

This Study

Question: Do economic incentives affect decisions to join religious groups?

- Focus on Seventh-Day Adventists (SDA), for whom certain crops are taboo (tobacco, coffee, and tea)
- Calculate a time-varying and location-specific “cost of being SDA”; relate this to variation in new SDA memberships in Africa

Identification: Interaction of...

- (Time-varying) annual prices of crops that are taboo to SDA
- (Time-invariant) crop suitability of locations

Data:

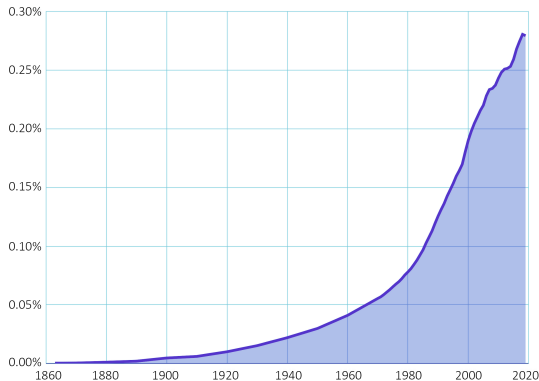
- FAO: 1) crop suitability of locations, 2) annual crop prices
- SDA online database of new membership by year and location

- Religious groups differ in their characteristics (e.g., structures, taboos, location)
- People differ in their characteristics, as well as their “match” with different religious groups
 - They choose the religious group with the highest match value
- In equilibrium, multiple religious groups can have nonzero membership in a location
- Exogenous changes in costs and benefits can affect decisions to join religious groups (by changing match values)
 - In particular: increases in the cost of membership in a particular religious group should reduce membership in that group

Seventh-Day Adventists (SDA)

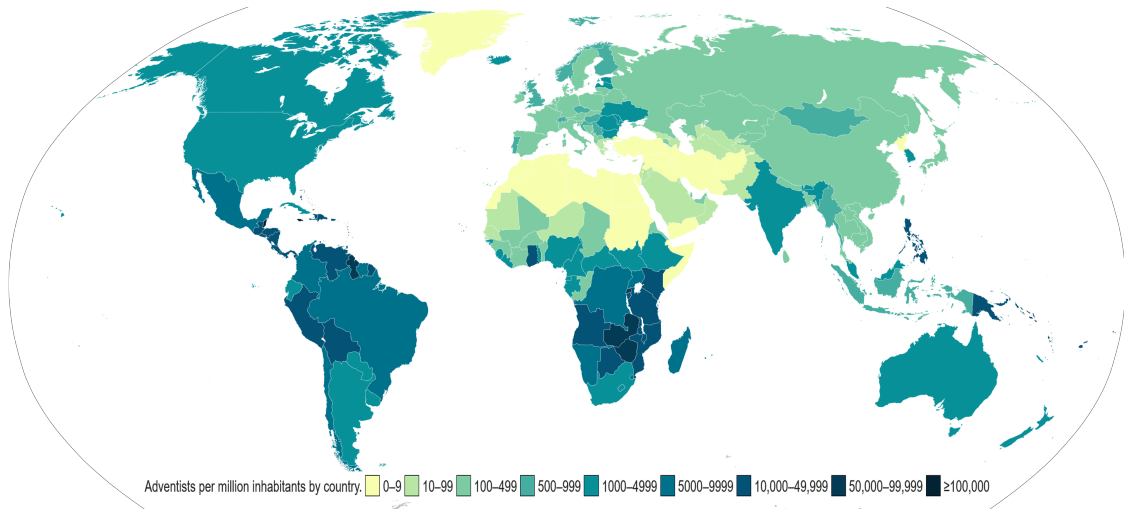
- Originated in Battle Creek, MI in 1872
- Widespread missionary outreach
 - One of the world's fastest-growing religious groups, primarily from growth in developing countries
- Roughly 21 million members worldwide in 2019 (0.27% of world population)
 - Substantial presence in Sub-Saharan Africa (one-third of worldwide membership)
 - In southern Africa, 1.8% of population is SDA

SDA membership, % of world population



Source: https://en.wikipedia.org/wiki/Seventh-day_Adventist_Church

Adventists around the world



Map from https://en.wikipedia.org/wiki/Seventh-day_Adventist_Church

Adventist crop taboos

Adventists have long placed a high value on healthy living.

From Ellen White, one of the founders of the SDA church:

“Those who use tobacco, tea, and coffee should lay aside those idols, and put their cost into the treasury of the Lord.”

- Adventists are **strongly discouraged** from consuming tobacco, coffee, and tea
- The taboo extends to **production** of these crops

Source: <https://www.fulcrum7.com/blog/2020/5/18/coffee-the-christian-and-the-church>

Key RHS (causal) variable: $SDAcost_{it}$

- Economic cost of abiding by SDA crop taboos at location i , time t
- For each pixel (location) i at time t , construct:
 - $R_{SDA,it}$: Maximized agricultural revenue, if constrained by SDA crop taboos
 - $R_{max,it}$: Maximized agricultural revenue, if unconstrained

We calculate $R_{SDA,it}$ and $R_{max,it}$ at pixel level, considering:

- pixel i 's agricultural productivity across 37 crops (in kg. per hectare)
- crop price per kg. in year t

Take the difference between these two:

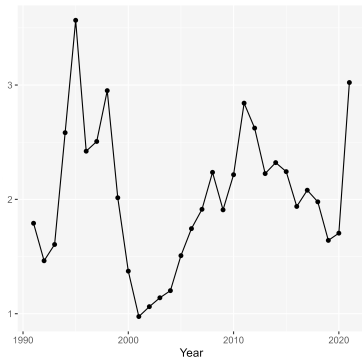
- $SDAcost_{it} = R_{max,it} - R_{SDA,it}$

With these pixel-level data, we can readily aggregate to higher geographic levels

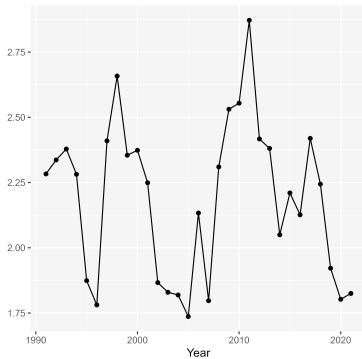
Crops considered: banana, barley, buckwheat, phaseolous bean, cabbage, carrot, cassava, chickpea, coconut, cacao, **coffee**, cotton, cowpea, dry peas, flax fibre, gram, groundnut, maize, millet, oat, oil palm, olive, onion, pigeonpea, rapeseed, dryland rice, rye, sorghum, soybean, sweet potato, sugar beet, sugar cane, sunflower, **tea**, **tobacco**, tomato, wheat, yam

Price Variation

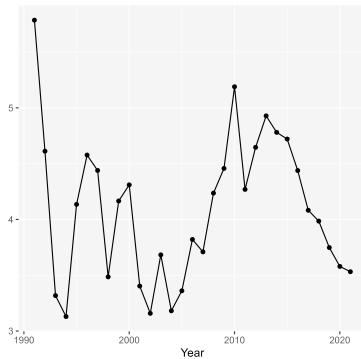
FAO Comtrade Prices for Taboo Crops, 2012 USD per Kg



Coffee



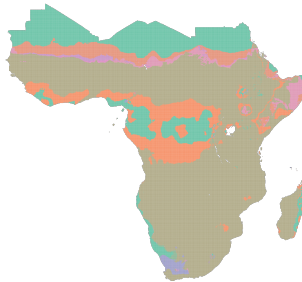
Tea



Tobacco

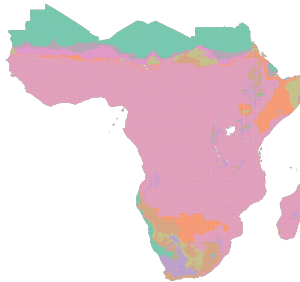
Top Revenue-Generating Crop

Top Revenue-Generating Crop by Pixel in 1995



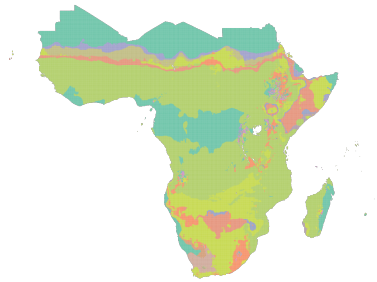
1995

Top Revenue-Generating Crop by Pixel in 2011



2011

Top Revenue-Generating Crop by Pixel in 2020

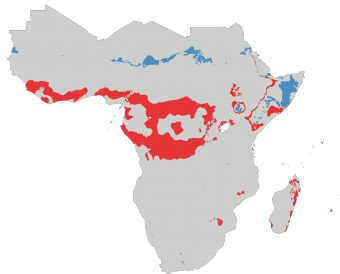


2020

Top Revenue-Generating Crop Taboo for SDA

Top Revenue-Generating Crop in 1995 Prohibited by SDA

■ Not Prohibited ■ coffee ■ tobacco



1995

Top Revenue-Generating Crop in 2011 Prohibited by SDA

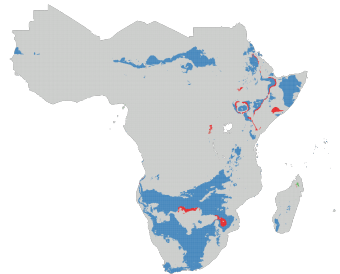
■ Not Prohibited ■ coffee ■ tobacco ■ tea



2011

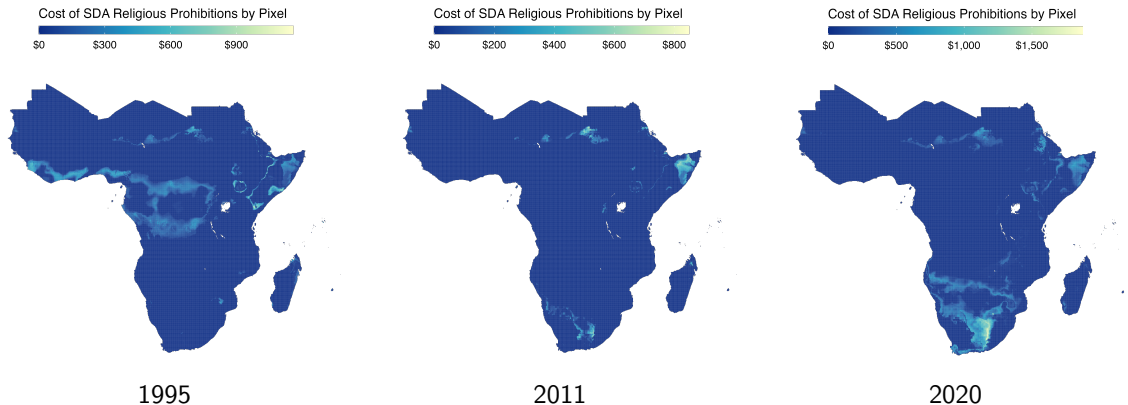
Top Revenue-Generating Crop in 2020 Prohibited by SDA

■ Not Prohibited ■ coffee ■ tobacco ■ tea



2020

Pixel-level SDA_{cost}_{it} (Cost of Being an SDA Member)



SDA Data: Adventist Office of Archives, Statistics, and Research

- Detailed membership data for all sub-national administrative divisions, dating back to before 1900.
- Include a list of districts/regions.

Michigan Conference | Yearly Statistics (1932-2021)
(Back to Field Summary)

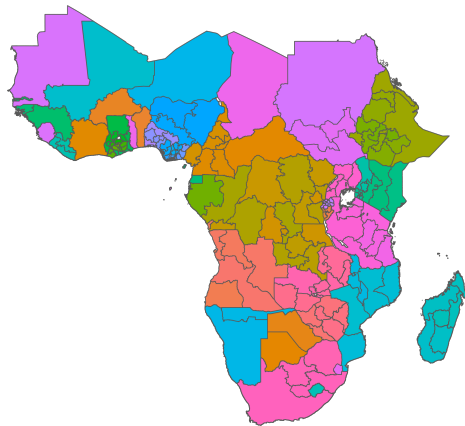
Click for chart, and for description

Year	Churches	Companies	Beginning Membership	Baptisms	Former Member Baptisms	Professions of Faith	Transfers In	Transfers Out	Deaths	Dropped	Ministries	Total Gains	Total Losses	Adjust	Net Growth	Ending Membership	Growth Rate	Accession Rate	Death Rate	Ordained Ministers	Licensed Ministers
1932	147			835		426						1285	0		9497					31	8
1933	149		9497	655		87						740	0		343	9842	3.63%	7.79%	0.00%	32	8
1934	150		9842	658		98						734	0		408	10242	4.00%	7.68%	0.00%	32	8
1935	153		10242	519		84						603	0		248	10490	2.42%	5.89%	0.00%	27	8
1936	153		10490	450		78						488	0		-40	10450	-0.38%	4.61%	0.00%	27	8
1937	150		10450	443		63						506	0		143	10593	1.37%	4.84%	0.00%	31	7
1938	152		10593	555		58						605	0		196	10789	1.85%	5.72%	0.00%	27	7
1939	157		10789	647		58						765	0		168	10949	1.48%	6.53%	0.00%	27	7
1940	161		10949	574		68						842	0		308	11249	2.74%	5.88%	0.00%	38	13
1941	147		11249	644		57						791	0		269	11538	2.59%	6.23%	0.00%	33	10
1942	150		11538	795		69						774	0		383	11999	3.91%	6.12%	0.00%	37	20
1943	149		11999	537		32						589	0		85	11984	0.11%	4.99%	0.00%	36	20
1944	149		11984	488		38						538	0		232	12238	1.94%	4.49%	0.00%	41	23
1945	143		12238	482		49						531	0		-848	11568	-6.94%	4.31%	0.00%	36	22
1946	142			634		43						677	0			11674				34	28
1947	141			10674		35						742	0		353	12027	3.02%	6.38%	0.00%	37	25
1948	144		12027	636		73						799	0		-57	11970	-0.47%	5.96%	0.00%	42	20
1949	147		11970	636		58						892	0		239	12209	2.00%	5.78%	0.00%	46	18
1950	150		12209	655		79						712	0		313	12522	2.58%	5.83%	0.00%	49	24
1951	152		12522	644		34						676	0		222	12744	1.77%	5.41%	0.00%	49	34
1952	154		12744	617		35						672	0		305	13049	2.39%	5.27%	0.00%	54	42
1953	159		13049	590		35						645	0		239	13308	1.98%	4.94%	0.00%	58	49
1954	159		13308	891		27						830	0		534	13842	4.05%	6.24%	0.00%	62	62
1955	159		13842	624		47						871	0		299	14141	2.16%	4.85%	0.00%	79	40
1956	161		14141	663		49						712	0		251	14372	1.63%	5.04%	0.00%	81	40
1957	162		14372	782		32						794	0		394	14766	2.74%	5.57%	0.00%	85	40
1958	159		14766	885		43						908	0		308	15272	3.43%	6.13%	0.00%	79	35
1959	161		15272	719		42						781	0		218	15482	1.38%	4.98%	0.00%	82	30
1960	162		15482	831		37						888	0		333	16015	3.44%	5.74%	0.00%	88	38
1961	163		16015	1217		47						1264	0		703	16718	4.39%	7.89%	0.00%	82	37
1962	165		16718	1077		39						1136	0		518	17237	3.10%	6.88%	0.00%	86	37
1963	164		17237	869		19						902	0		318	17596	2.08%	5.37%	0.00%	96	24
1964	162		17596	735		42						777	0		255	17821	1.38%	4.42%	0.00%	88	33
1965	162		17821	658		32	1826	1811	255	299	29	2511	2385		126	17947	0.71%	3.94%	1.43%	95	30
1966	162		17947	657			1701	1430	210	413		2538	2855		283	18239	1.57%	5.33%	1.17%	103	32
1967	160		18239	825			2082	1946	236	371		2865	2553		352	18582	1.87%	4.51%	1.29%	96	27
1968	161		18582	845			1753	1744	247	342		2894	2233		361	18943	1.94%	4.54%	1.33%	96	29

ASTR

Geographic Units: Locations

- Match SDA divisions with administrative ones determine borders: *SDA Locations*
- Current Data: 1991-2021
- Location definitions have breaks over time: we follow each separately
- N: 99 - 193



rSDA Units: 2020

Cost of being an SDA member by Location

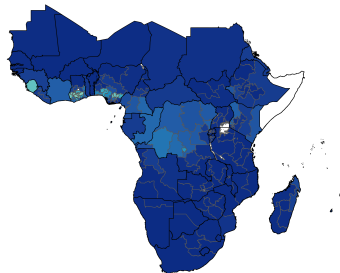
Cost of SDA Prohibitions by rSDA in 1995



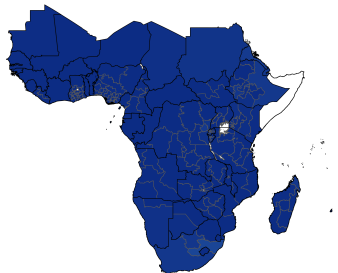
Cost of SDA Prohibitions by rSDA in 2011



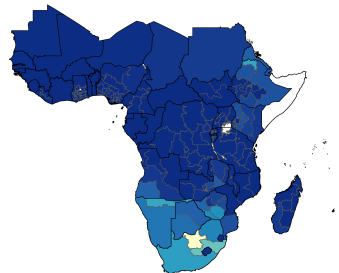
Cost of SDA Prohibitions by rSDA in 2020



1995



2011



2020

Summary Statistics

	Mean	SD	p10	p25	p50	p75	p90	Min	Max	N
Costs:										
$SDAcost_{ijt}$	20.21	72.08	0.00	0.00	0.00	1.60	45.59	0.00	764.93	4,046
Dependent Variables:										
TotalGains	3,750	7,342	100	400	1,385	4,788	9,615	0	150,624	4,077
Baptisms	2,974	4,237	91	357	1,214	4,053	7,953	0	46,947	4,077

Regression Specification

$$\log(y_{ijt}) = \beta_0 + \beta_1 \log(SDAcost_{ij|t,t-k}) + \nu_i + \phi_t + \theta_{jt} + \epsilon_{ijt} \quad (1)$$

- y_{ijt} : outcome variable (e.g., SDA baptisms) in location i , country j , year t
- $SDAcost_{ij|t,t-k}$: average cost of SDA crop taboos in location i , country j , years t through $t - k$
- ν_i : location fixed effects
- ϕ_t : year fixed effects
- θ_{jt} : country-year fixed effects

Standard errors clustered by country j

Regression Results: Total Membership Gains

	log(Total Gains)					
	(1)	(2)	(3)	(4)	(5)	(6)
$\log(SDAcost_{t,t-1})$	-0.062*** (0.019)			-0.026** (0.012)		
$\log(SDAcost_{t,t-2})$		-0.074*** (0.020)			-0.035** (0.017)	
$\log(SDAcost_{t,t-3})$			-0.065*** (0.022)			-0.032** (0.014)
Observations	3726	3726	3726	3065	3065	3065
Adjusted R^2	0.837	0.837	0.837	0.873	0.873	0.873
Mean Dep. Var.	7.087	7.087	7.087	7.564	7.564	7.564
# of Locations	380	380	380	352	352	352
Country x Year FE	N	N	N	Y	Y	Y

Standard errors clustered by country in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Regression Results: Baptisms

	log(Baptisms)					
	(1)	(2)	(3)	(4)	(5)	(6)
$\log(SDAcost_{t,t-1})$	-0.049** (0.018)			-0.032** (0.012)		
$\log(SDAcost_{t,t-2})$		-0.057*** (0.020)			-0.037** (0.014)	
$\log(SDAcost_{t,t-3})$			-0.046** (0.022)			-0.027* (0.014)
Observations	3726	3726	3726	3065	3065	3065
Adjusted R^2	0.843	0.843	0.842	0.882	0.882	0.882
Mean Dep. Var.	6.938	6.938	6.938	7.401	7.401	7.401
# of Locations	380	380	380	352	352	352
Country x Year FE	N	N	N	Y	Y	Y

Standard errors clustered by country in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Religious “Market” Concentration

Are the effects of $SDAcost_{it}$ heterogeneous wrt religious market structure?

- World Religion Database (2000): Calculate Herfindahl-Hirschman Index by country

$$\log(y_{ijt}) = \alpha_0 + \alpha_1 \log(SDAcost_{ij|t,t-k}) + \alpha_2 \log(SDAcost_{ij|t,t-k}) \cdot HHI_{2000} + \nu_i + \phi_t + \theta_{jt} + \epsilon_{ijt} \quad (2)$$

HH_{2000} : higher value $\Rightarrow$ less *competition*

Heterogeneity: Total Membership Gains

	log(Total Gains)					
	(1)	(2)	(3)	(4)	(5)	(6)
$\log(SDAcost_{t,t-1})$	-0.000 (0.034)			0.085** (0.035)		
$\log(SDAcost_{t,t-1}) \cdot HH_{2000}$	-0.182* (0.091)			-0.330*** (0.099)		
$\log(SDAcost_{t,t-2})$		-0.004 (0.039)			0.066 (0.060)	
$\log(SDAcost_{t,t-2}) \cdot HH_{2000}$		-0.207* (0.107)			-0.301* (0.171)	
$\log(SDAcost_{t,t-3})$			0.013 (0.038)			0.042 (0.061)
$\log(SDAcost_{t,t-3}) \cdot HH_{2000}$			-0.234** (0.106)			-0.229 (0.182)
Observations	3726	3726	3726	3065	3065	3065
Adjusted R^2	0.837	0.838	0.837	0.873	0.873	0.873
# of Locations	380	380	380	352	352	352
Country x Year FE	N	N	N	Y	Y	Y

Standard errors clustered by country in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Heterogeneity: Baptisms

	log(Baptisms)					
	(1)	(2)	(3)	(4)	(5)	(6)
$\log(SDAcost_{t,t-1})$	0.006 (0.033)			0.100*** (0.032)		
$\log(SDAcost_{t,t-1}) \cdot HH_{2000}$	-0.162* (0.081)			-0.392*** (0.097)		
$\log(SDAcost_{t,t-2})$		0.009 (0.038)			0.106** (0.039)	
$\log(SDAcost_{t,t-2}) \cdot HH_{2000}$		-0.196* (0.098)			-0.429*** (0.120)	
$\log(SDAcost_{t,t-3})$			0.028 (0.038)			0.095** (0.041)
$\log(SDAcost_{t,t-3}) \cdot HH_{2000}$			-0.221** (0.099)			-0.377*** (0.131)
Observations	3726	3726	3726	3065	3065	3065
Adjusted R^2	0.843	0.843	0.843	0.882	0.882	0.882
# of Locations	380	380	380	352	352	352
Country x Year FE	N	N	N	Y	Y	Y

Standard errors clustered by country in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Conclusion and Next Steps

- People do respond to economic costs when making decisions to join a religious group
- This may be the first such evidence in economics

Next steps:

- Further explore heterogeneity in effects of $SDAcost_{it}$: wrt religious market concentration, % SDA in locality, etc.
 - Write down a model that generates predictions about heterogeneity in effects of $SDAcost_{it}$
- Expand to other continents?